

Customer Behavior Analysis

1. The Business Problem

The primary objective was to help a retail company understand customer purchasing behavior to drive revenue, improve retention, and optimize marketing. Key challenges addressed included:

- Identifying high-revenue product categories and seasonal demand.
- Understanding the impact of discounts on actual purchase value.
- Segmenting customers (New vs. Loyal) to move away from generic marketing.

2. Data Cleaning with Python

Using **Pandas**, I transformed a messy raw dataset into a clean, analysis-ready format. Key cleaning steps included:

- **Logical Mapping:** I standardized product categories by mapping specific items (e.g., "T-shirt" to "Clothing", "Laptop" to "Electronics") using a custom dictionary to fix mixed or incorrect entries.
- **Missing Value Imputation:**
 - For **Size**, I applied logic-based filling: "Not Applicable" for Electronics/Footwear and "Free Size" for Accessories.
 - For **Review Ratings** and **Purchase Amounts**, I filled nulls using the **median** value per specific item to maintain statistical integrity.
- **Deduplication:** I identified and removed 50 duplicate records based on `customer_id` to ensure unique transaction reporting.
- **Database Prep:** Renamed columns to lowercase with underscores (e.g., `purchase_amount`) and established a connection via `sqlalchemy` to export the cleaned data directly into a **MySQL** database.

3. Exploratory Data Analysis (EDA) in SQL

I used SQL to perform a deep dive into the business questions, moving beyond simple counts to actionable insights:

- **Revenue Leaders:** Discovered that **Electronics** leads revenue with over \$485k, followed by Accessories.
- **Discount Effectiveness:** Used subqueries to identify "Premium Discount Users"—customers who used a discount but still spent above the average purchase amount.
- **Customer Segmentation:** Created a segmentation model using CASE statements:
 - **New Customers:** 0 previous purchases (561 users).
 - **Returning:** 1-10 purchases (902 users).
 - **Loyal:** >10 purchases (3,537 users).
- **Category Deep Dives:** Used **Window Functions** (RANK() OVER) to find the top 3 most purchased items within every category.

4. Visualization in Power BI

The final solution involved building an interactive dashboard to make these insights digestible for stakeholders:

- **Strategic Health KPIs:** Tracked Total Revenue (\$965.81K), Total Customers (5K), and Average Review Rating (3.61).
- **Behavioral Analytics:**
 - **Demographic Breakdown:** Visualized revenue by Gender and Age Groups (revealing the 51+ group as a major revenue driver).
 - **Shipping & Payment:** Created bar charts comparing revenue across shipping types (Standard vs. Express) and payment methods (Debit Card leads at \$0.25M).

- **Performance Rankings:** Developed "Top 5" and "Bottom 5" charts for products by both ratings and revenue to guide inventory and quality control.